22

Measuring Employment and Unemployment

The country is hungry for information; everything of a statistical character, or even a statistical appearance, is taken up with an eagerness that is almost pathetic; the community have not yet learned to be half skeptical and critical enough in respect to such statements.

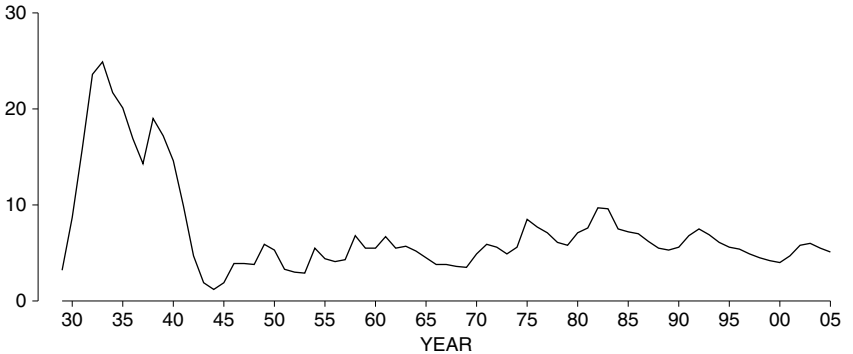
—GENERAL FRANCIS A. WALKER, SUPERINTENDENT OF THE 1870 CENSUS

1. INTRODUCTION

The unemployment rate is one of the most important numbers published by the government. Unemployment was only 3% in 1929, before the stock market crash (figure 1 on the next page). It reached 25% in the depths of the Depression, and remained fairly high until the U.S. entered World War II. More recently, as a result of anti-inflationary practices adopted by the Federal Reserve Board in 1981, the economy went into a deep recession in 1982–83, and the unemployment rate nearly reached 10%. By the late 1980s, the rate had dropped below 6%; in many metropolitan areas, there were shortages of skilled workers. In 2003, after the Internet bubble collapsed, the rate returned to 6%. Unemployment declined from there to the end of 2005.

The government agency in charge of the employment numbers is the Bureau of Labor Statistics. But how do they know who is employed or unemployed? Employment statistics are estimated from a sample survey—the Current Population Survey. This massive and beautifully organized sample survey is conducted

Figure 1. The unemployment rate from 1929 to 2005.



Source: *Employment and Earnings*, January 1976, table A-1; July 1989, table A-3; December 2005, table A-1.

monthly for the Bureau of Labor Statistics by the Census Bureau.[1] During the week containing the 19th day of the month, a field staff of 1,700 interviewers canvasses a nationwide probability sample of about 110,000 people. The size of the labor force, the unemployment rate, and a host of other economic and demographic statistics (like the distribution of income and educational level) are estimated from survey results, at a cost which in 2005 was about $60 million a year. The results are published in:

- *Monthly Labor Review*,
- *Employment and Earnings* (monthly),
- *The Employment Situation* (monthly),
- *Current Population Reports* (irregular),
- *Statistical Abstract of the United States* (annual),
- *Economic Report of the President* (annual).

The object of this chapter is to present the Current Population Survey in detail, from the ground up. This will illustrate and consolidate the ideas introduced in previous chapters. It should also make other large-scale surveys easier to understand. The main conclusions from this case study:

- In practice, fairly complicated probability methods must be used to draw samples. Simple random sampling is only a building-block in these designs.
- The standard-error formulas for simple random samples do not apply to these complicated designs, and other methods must be used for estimating the standard errors.

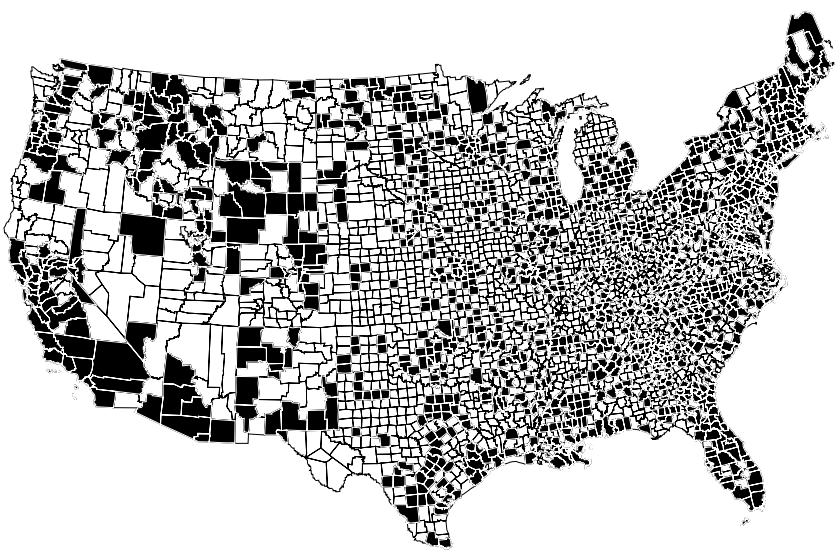
2. THE DESIGN OF THE CURRENT POPULATION SURVEY

The Current Population Survey is redesigned periodically by the Census Bureau, to take advantage of new information and to accomplish new objectives. There was a major redesign in the early 2000s, using data from the 2000 census.

There were 3,142 counties and independent cities in the U.S. As the first step in the redesign process, the Bureau put these together into groups to form 2,025 *Primary Sampling Units* (or PSUs, for short). Each PSU consisted either of a city, or a county, or a group of contiguous counties.[2] These PSUs were sorted into 824 *strata*, chosen so the PSUs in each stratum would resemble each other on certain demographic and economic characteristics (like unemployment at the time of stratification, the number of large households, and the number of workers in retail trade). The strata do not cross state lines. Many of the larger PSUs, like New York or Los Angeles, were put into strata by themselves.

The sample was chosen in two stages. To begin with, one PSU was chosen from each stratum, using a probability method which ensured that within the stratum, the chance of a PSU getting into the sample was proportional to its population. Since there were 824 strata, the first stage resulted in a sample of 824 PSUs. Until the next redesign (after the 2010 census), all interviewing for the Survey takes place in these 824 PSUs and in no others. The PSUs for an earlier design are shown in figure 2.

Figure 2. Primary Sampling Units for the Current Population Survey: the 1995 sample design with 792 PSUs.



Note: Alaska and Hawaii not shown.
Source: Bureau of the Census, Statistical Methods Division.

Each PSU was divided up into *Ultimate Sampling Units* (or USUs), consisting of about 4 housing units each. At the second stage, some USUs were picked at random for the sample. In the end, every person age 16 and over living in a selected USU in a selected PSU gets into the Current Population Survey. For the U.S. as a whole, the sampling rate is about 1 in 2,000. But the rate varies from about 1 in 300 for D.C. or Wyoming to 1 in 3,000 for large states like Califor-

nia, New York, and Texas.[3] The objective is to estimate unemployment rates in each of the 50 states and the District of Columbia with about the same precision.[4] This meant equalizing, at least roughly, the absolute sizes of the 51 subsamples (section 4 of chapter 20). So the ratio of sample size to population size has to be different from state to state.

The Bureau's choices for the sample to be used from 2005 to 2015 were all made well before 2005. The design even provided for people who were going to live in housing yet to be constructed. And in fact, the Bureau chose not just one sample but 16 different ones, in order to rotate part of the sample every month. After it gets into the sample, a housing unit is kept there for 4 months, dropped out for 8 months, and then brought back for a final 4 months. Why rotate the sample? For one reason, the interviewers may wear out their welcome after a while. Besides that, people's responses probably change as a result of being interviewed, progressively biasing the sample (this is called *panel bias*). For instance, there is some evidence to show that people are more likely to say they are looking for a job the first time they are interviewed than the second time. Then why not change the sample completely every month? Keeping part of it the same saves a lot of money. Besides that, having some overlap in the sample makes it easier to estimate the monthly changes in employment and unemployment.

3. CARRYING OUT THE SURVEY

The Survey design for 2005 produces 72,000 housing units to be canvassed each month. Of these, about 12,000 are ineligible for the sample (being vacant, or even demolished since the sample was designed). Another 4,500 or so are unavailable, because no one is at home, or because those at home will not cooperate. That leaves about 55,500 housing units in the Survey. All persons age 16 or over in these housing units are asked about their work experience in the previous week. On the basis of their answers, they are classified as:

- employed (those who did any paid work in the previous week, or were temporarily absent from a regular job);
- or unemployed (those who were not employed the previous week but were available for work and looking for work during the past four weeks);
- or outside the labor force (defying Aristotle, the Bureau defines this as the state of being neither employed nor unemployed).[5]

The employed are asked about the hours they work and the kind of job they have. The unemployed are asked about their last job, when and why they left it, and how they are looking for work. Those outside the labor force are asked whether they are keeping house, or going to school, or unable to work, or do not work for some other reason (in which case, they are asked to specify what). Results for November 2005 are shown in table 1.

Table 1. The civilian non-institutional population age 16 and over.[6] Bureau of Labor Statistics estimates, November 2005. In millions.

Employed	142.97		
Unemployed	7.27		
Labor force		150.24	
Outside the labor force		76.96	
Total			227.20

Source: *Employment and Earnings*, December 2005, table A-13.

By definition, the *civilian labor force* consists of the civilians who are either employed or unemployed. In November 2005, that amounted to $142.97 + 7.27 = 150.24$ million people.[7] The *unemployment rate* is the percentage of the civilian labor force which is unemployed, and that came to

$$\frac{7.27}{150.24} \times 100\% \approx 4.8\%.$$

This 4.8% is an average rate of unemployment, over all the subgroups of the population. Like many averages, it conceals some striking differences. These differences are brought out by a process of cross-tabulation. Unemployment falls more heavily on teenagers and blacks, as shown by by table 2.

Table 2. Unemployment rates by race, age, and sex. Bureau of Labor Statistics estimates, November 2005. In percent.

		Age group		
Race	*Sex*	*16–19*	*20–64*	*65 and over*
White	Male	15.1	3.5	3.2
White	Female	12.4	3.7	2.4
Black	Male	41.6	9.6	6.8
Black	Female	31.7	9.0	4.5

Source: *Employment and Earnings*, December 2005, table A-13.

The overall unemployment rate is quite variable, as shown in figure 1 on p. 396. But the pattern of rates in table 2 is quite stable in certain respects. For instance, the unemployment rate for blacks has been roughly double the unemployment rate for whites over the period 1961–2005. One development is worth noting. From the 1990s onwards, unemployment rates for men have become higher than those for women—a change from the past.

Unemployment numbers are published for much finer classifications than the ones shown in table 2, including marital status, race, age, sex, type of last job, reason for unemployment (for instance, fired or quit), and duration of unemployment. The Bureau starts with a huge sample. But by the time it comes down to the white men age 35–44 who quit managerial jobs, have been out of work for 5 to 14 weeks, and are looking for work by reading the newspapers, there might not be

Figure 3. Table A-32, *Employment and Earnings*, December 2005.

HOUSEHOLD DATA
NOT SEASONALLY ADJUSTED

nemployed persons by reason for unemployment, sex, and age

in thousands)

Reason	Total, 16 years and over		Men, 20 years and over		Women, 20 years and over		Both sexes, 16 to 19 years	
	Dec. 2004	Dec. 2005	Dec. 2004	Dec. 2005	Dec. 2004	Dec. 2005	Dec. 2004	Dec. 2005
NUMBER OF UNEMPLOYED								
unemployed	7,599	6,956	3,727	3,355	2,802	2,707	1,070	894
ers and persons who completed temporary jobs	4,166	3,622	2,573	2,212	1,433	1,281	160	129
nporary layoff	1,040	1,013	709	679	255	287	77	47
temporary layoff	3,126	2,609	1,864	1,534	1,178	993	84	82
anent job losers	2,272	1,866	1,302	1,072	908	743	61	51
ons who completed temporary jobs	854	743	562	461	270	250	23	31
vers	845	752	398	339	369	335	78	78
ants	2,040	2,083	683	722	894	1,000	462	361
trants	548	499	74	82	105	91	369	325
PERCENT DISTRIBUTION								
unemployed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
ers and persons who completed temporary jobs	54.8	52.1	69.0	65.9	51.2	47.3	15.0	14.4
mporary layoff	13.7	14.6	19.0	20.2	9.1	10.6	7.2	5.3
n temporary layoff	41.1	37.5	50.0	45.7	42.1	36.7	7.8	9.2
vers	11.1	10.8	10.7	10.1	13.2	12.4	7.3	8.8
ants	26.8	30.0	18.3	21.5	31.9	36.9	43.2	40.4
ntrants	7.2	7.2	2.0	2.4	3.8	3.4	34.5	36.4
UNEMPLOYED AS A PERCENT OF THE CIVILIAN LABOR FORCE								
ers and persons who completed temporary jobs	2.8	2.4	3.4	2.9	2.2	1.9	2.3	1.9
vers	.6	.5	.5	.4	.6	.5	1.1	1.1
ants	1.4	1.4	.9	.9	1.4	1.5	6.7	5.3
ntrants	.4	.3	.1	.1	.2	.1	5.4	4.8

Beginning in January 2005, data reflect revised population controls used in the household survey.

too many cases left. Figure 3 shows estimates by reason for unemployment, sex, and age. (*Employment and Earnings* goes into much more detail.)

In general, by the time a large sample is cross-tabulated, there will be only very small subsamples in some classifications. Inferences about the corresponding subpopulations would be quite uncertain. Now, suppose that each estimate is within 1% of its true value with probability 95%, say. With a thousand estimates (which is about the number in *Employment and Earnings*), it would not be surprising if a few of them are quite a bit more than 1% off. The Bureau takes a big sample because it has to make many estimates about many subpopulations, and it wants to be reasonably confident that they are all fairly accurate. In fact, the Bureau will not make estimates when a subsample drops below a threshold size of about 50 cases.

"The dip in sales seems to coincide with the decision to eliminate the sales staff."

Drawing by Leo Cullum; © 2006 The New Yorker Magazine, Inc.

4. WEIGHTING THE SAMPLE

Suppose that one month, in the Bureau's sample of 110,000 people, there are 3,836 who are unemployed. The Bureau is sampling 1 person in 2,000 from the civilian non-institutional population age 16 and over. So it is natural to think that each person in the sample represents 2,000 people in the country. Then the way to estimate the total number of unemployed in the population is to weight up the sample number of 3,836 by the factor of 2,000:

$$2{,}000 \times 3{,}836 = 7{,}672{,}000$$

However, the Bureau does not do anything that simple. Not everybody in the sample gets the same weight. Instead, the Bureau divides the sample up into groups (by age, sex, race, and area of residence) and weights each group up separately.

There is a good reason for all the complexity. The sampling rate is different from one stratum to another, and the weights have to compensate; otherwise, the estimates could be quite biased. Moreover, the weights are used to control the impact of chance variation. For example, suppose there are too many white males age 16–19 in the sample, relative to their share in the total population. Unemployment is high in this group, which would make the overall unemployment rate in the sample too high. The Bureau has a fix: any group which is over-represented in the sample gets proportionately smaller weights, bringing the sample back into line with the population. On the other hand, if a group is under-represented, the weights are increased. Adjusting the weights this way helps to correct imbalances caused by chance variation. That reduces sampling error.[8]

5. STANDARD ERRORS

In estimating the unemployment rate, precision counts. For example, a definite picture of the economy is given by saying that the unemployment rate is 7.0% ± 0.1 of 1%. However, 7% ± 3% covers everything from boom to bust. So, it is important to know how good the estimates really are. Procedures we have discussed in previous chapters do not apply, because the Bureau is not using a simple random sample. In particular, at the second stage of its sampling procedure the Bureau chooses some ultimate sampling units (USUs). A USU is a *cluster* of about four adjacent housing units. Every person age 16 and over living in one of these USUs gets into the sample (section 2). A cluster is all or nothing: either everybody in the cluster gets into the sample, or nobody does. People living in the same cluster tend to be similar to one another in many ways. Information about each one says something about all the others, in terms of family background, educational history, and employment status.

With simple random sampling, by comparison, if one person in the cluster gets into the sample, the other people still have only a small chance of getting in. As a result, each person drawn into a simple random sample provides additional information, independent of the persons drawn previously. The Bureau's cluster sample of 110,000 persons contains less information than a simple random sample of the same size: cluster samples involve a lot of redundancy. Thus, clustering tends to reduce the precision of the Bureau's estimates. On the other hand, the weights improve precision. All in all, computing SEs for the Bureau's estimates is a delicate business.

As it turns out, with a cluster sample the standard errors can themselves be estimated very closely from the data, using the *half-sample method*. Although the details are complicated and take a lot of computer power, the idea is simple. If the Bureau wanted to see how accurate the Current Population Survey was, one thing to do would be make another independent survey following exactly the same procedures. The difference between the two surveys would give some idea of how reliable each set of results was.

Nobody would seriously propose to replicate the Current Population Survey, at a cost of another $60 million a year, just to see how reliable it is. But the Bureau can get almost the same effect by splitting the Survey into two independent pieces which have the same chance behavior (hence the name, "half-sample method"). Suppose for instance that one piece of the survey estimates the civilian labor force at 150.5 million, and the other comes in at 150.7 million. This difference is due to chance error. The pooled estimate of the civilian labor force is

$$\frac{150.5 + 150.7}{2} = 150.6 \text{ million}$$

The two individual estimates are 0.1 million away from their average, and the standard error is estimated by this difference of 0.1 million.

Of course, an estimated standard error based on only one split may not be too reliable. But there are many different ways to split the sample. The Bureau looks at a number of them and combines the standard errors by taking the root-mean-square. This completes the outline of the half-sample method.[9] Some of the estimated standard errors for November 2005 are shown in table 3.

Table 3. Estimated standard errors, November 2005.

	Estimate	*Standard error*
Civilian labor force	150.24 million	300,000
Employment	142.97 million	323,000
Unemployment	7.27 million	155,000
Unemployment rate	4.8%	0.1 of 1%

Source: *Employment and Earnings*, December 2005, tables A13, 1B, and 1C.

How do the estimated standard errors in table 3 compare to those for a simple random sample of the same size and composition? Calculations show that for estimating the size of the labor force, the Bureau's standard error is about 8% smaller than that for a simple random sample: the weights are doing a good job. For estimating the number of unemployed, however, the Bureau's sample is about 30% worse than a simple random sample: the clustering hurts.[10]

So why doesn't the Bureau use simple random sampling? For one thing, there is no list showing all the people age 16 and over in the U.S., with current addresses. Even if there were such a list, taking a simple random sample from it would produce people spread thinly throughout the country, and the cost of interviewing them would be enormous. With the Bureau's procedure, the sample is bound to come out in clumps in relatively small and well-defined areas, so the interviewing cost is quite manageable. In 2005, this was about $100 per interview. The Bureau's sample design turns out to be amazingly cost effective.

The comparison between the Bureau's design and a simple random sample points to a real issue. To compute a standard error properly, you need more than the sample data. You need to know how the sample was picked. With a simple random sample, there is one SE. With a cluster sample, there is another. The formulas which apply to simple random samples will usually underestimate the standard errors in cluster samples. (These issues came up before, in the context of the Gallup Poll: sections 4 and 5 of chapter 21.)

> Cluster samples are less informative than simple random samples of the same size. So the simple random sample formulas for the standard error do not apply.

Exercise Set A

1. One month, the Current Population Survey sample amounted to 100,000 people. Of them, 62,000 were employed, and 3,000 were unemployed. True or false, and explain:
   (a) 65% of the sample was in the labor force.
   (b) The Bureau would estimate that 65% of the population was in the labor force.

2. The Current Population Survey sample is split into two independent halves. From one half, the number of employed persons is estimated as 151.5 million; from the other, it is estimated as 151.3 million. Combine these two estimates, and attach a standard error to the result.

3. (Hypothetical.) The Health Department in a certain city takes a simple random sample of 100 households. In 80 of these households, all the occupants have been vaccinated against polio. So the Department estimates that for 80% of the households in that city, all the occupants have been vaccinated against polio. With this information, can you put a standard error on the 80%? Find the SE, or explain what other information is needed.

4. (Continues exercise 3.) The Department interviews every person age 25 and over in the sample households. They find 144 such persons, of whom 29 have college degrees. They estimate that 20.1% of the people age 25 and over in the city have college degrees. With this information, can you put a standard error on the 20.1%? Find the SE, or explain what other information is needed.

5. In election years, the Bureau makes a special report on voting, using the Current Population Survey sample. In 2000, about 55% of all the people of voting age in the sample said they voted; but only 52% of the total population of voting age did in fact vote.[11] Can the difference be explained as a chance error? If not, how else can it be explained? (You may assume that the Bureau's sample is the equivalent of a simple random sample of 75,000 people.)

6. In table 2 on p. 399, which estimate is more trustworthy: for white males age 20–64, or black males age 20–64? Explain briefly.

The answers to these exercises are on pp. A84–85.

6. THE QUALITY OF THE DATA

The data collected by the Survey are of very high quality: for many purposes, Survey data are considered to be more accurate than Census data. In any large-scale field operation, mistakes are inevitable. Since the Survey operates on a much smaller scale than the Census, it can afford better quality control. The key is careful selection, training, and supervision of the field staff. Interviewers are given about four days of training in survey procedures before they start work, and several hours a month of training while they are on the job. At least once a year, their work is observed by their supervisors. In addition, about 3% of the monthly sample (chosen by a separate probability sampling procedure) is reinterviewed by supervisors. All discrepancies are discussed with the interviewers. The interviewers' reports are *edited*, that is, checked for incomplete or inconsistent entries. For most items, error rates are low; and each error is reviewed with the person who made it.

7. BIAS

Bias is more insidious than chance error, especially if it operates more or less evenly across the sample. SEs computed by the half-sample method—or any other method—will not pick up that kind of bias. Measuring bias, even roughly, is hard work and involves going beyond the sample data.

> When bias operates more or less evenly across the sample, it cannot be detected just by looking at the data.

The Bureau has made unusually careful studies of the biases in the Current Population Survey. On the whole, these seem to be minor, although their exact sizes are not known. To begin with, the Survey design is based on Census data (section 2), and the Census misses a small percentage of the population. This percentage is not easy to pin down. Even if the Bureau knew it, they would still have a hard time adjusting the estimated number of unemployed (say) to compensate for the undercount, because the people missed by the Census are likely to be different from the ones the Census finds. A similar difficulty crops up in another place. The Survey misses about 10% of the people counted by the Census. To some extent, the weights bring these missing people back into the estimates. But non-response bias is not so easy to fix. The people missed by the Survey are probably different from the ones it finds, while the weights pretend they are the same.[12]

Next, the distinction between "employed" and "unemployed" is a little fuzzy around the edges. For example, people who have a part-time job but would like full-time work are classified as employed, but they really are partially unemployed. Moreover, people who want to work but have given up looking are classified as outside the labor force, although they probably should be classified as unemployed. The Bureau's criterion for unemployment, namely being without work, available for work, and looking for work, is necessarily subjective. In practice, it is a bit slippery. Results from the reinterview program (section 6) suggest the number of unemployed is higher than the Bureau's estimate, by several hundred thousand people. In this case, the bias is larger than the sampling error.[13] Over the period from 1980 to 2005, the number of unemployed has ranged from 5 to 10 million. Relatively speaking, both sampling error and non-sampling error are small.

8. REVIEW EXERCISES

Review exercises may cover previous chapters as well.

1. One month, there are 100,000 people in the Current Population Survey sample, of whom 63,000 are employed and 4,000 are unemployed.
   (a) True or false, and explain: the Bureau would estimate the percentage of the population who are unemployed as

$$\frac{4{,}000}{63{,}000 + 4{,}000} \times 100\% \approx 6\%$$

   (b) What happened to the other 33,000 people?

2. One month, there are 100,000 people in the Current Population Survey sample, and the Bureau estimates the unemployment rate as 6.0%. True or false, and explain: the standard error for this percentage should be estimated as follows—

$$\text{SE for number} = \sqrt{100{,}000} \times \sqrt{0.06 \times 0.94} \approx 75$$

$$\text{SE for percent} = \frac{75}{100{,}000} \times 100\% \approx 0.08 \text{ of } 1\%$$

3. One month, the Current Population Survey sample is split into two independent replicates. Using one replicate, the number of unemployed people is estimated as 7.1 million. The other replicate produces an estimate of 6.9 million. Using this information, estimate the number of unemployed people, and attach a standard error to the estimate.

4. Using the data in exercise 3, what can you say about the bias in the estimate?

5. A simple random sample is drawn at random _____ replacement. Options: with, without.

6. A box contains 250 tickets. Two people want to estimate the percentage of 1's in the box. They agree to use the percentage of 1's in 100 draws made at random from the box. Person A wants to draw with replacement; person B wants to draw without replacement. Which procedure gives a more accurate estimate? Or does it make any difference?

7. (Hypothetical.) A survey organization draws a sample of 100 households from 10,000 in a certain town, by the following procedure. First, they divide the town into 5 districts, with 2,000 households each. Then they draw 2 districts at random. Within each of the 2 selected districts, they draw 50 households at random.

   (a) Is this a probability sample?
   (b) Is this a simple random sample?

   Answer yes or no, and explain.

8. A supermarket chain has to value its inventory at the end of every year, and this is done on a sample basis. There is a master list of all the types of items sold in the stores. Then, auditors take a sample of the items and go through the shelves, finding the amounts in stock and prices for the sample items. To draw the sample, the auditors start by choosing a number at random from 1 to 100. Suppose this turns out to be 17. The auditors take the 17th, 117th, 217th, . . . items in the list for the sample. If the random number is 68, they take the 68th, 168th, 268th, . . . items. And so forth.

   (a) Is this a probability sample?
   (b) Is this a simple random sample?

   Answer yes or no, and explain.

9. As part of a study on drinking, the attitudes of a sample of alcoholics are assessed by interview.[14] Cases are assigned to interviewers at random. Some of the interviewers are teetotalers, others drink. Would you expect the two groups of interviewers to reach similar conclusions? Answer yes or no, and give reasons.

10. From "The Grab Bag" by L. M. Boyd in the *San Francisco Chronicle*: "The Law of Averages says that if you throw a pair of dice 100 times, the numbers tossed will add up to just about 683." Is this right? Answer yes or no, and explain.

11. A polling organization takes a simple random sample of 750 voters from a district with 18,000 voters. In the sample, 405 voters are for. Fill in the blanks, using the options below. Explain briefly,

    (a) The observed value of the _____ is 405.
    (b) The observed value of the _____ is 54%.
    (c) The expected value of the _____ is equal to the _____.

    Options:

    (i) number of voters in the sample who are for
    (ii) percentage of voters in the sample who are for
    (iii) percentage of voters in the district who are for

12. In 2004, there were 318,390 applications to buy guns in California.[15] A criminologist takes a simple random sample of 193 out of these applications, and finds that only 2 were rejected. True or false, and explain:

    (a) 2 out of 193 is 1.04%.
    (b) The SE on the 1.04% is 0.73%.
    (c) A 95%-confidence interval for the percentage of all 318,390 applications that were rejected is 1.04% $\pm$ 1.46%.

9. SUMMARY

1. Unemployment rates in the U.S. are estimated using the *Current Population Survey*.

2. This survey is based on a nationwide probability sample of about 110,000 persons, who are interviewed monthly. The design is more complicated than simple random sampling.

3. The Survey reweights the sample so it agrees with Census data on age, sex, race, state of residence, and certain other characteristics influencing employment status.

4. When a sample is taken by a probability method, it is possible not only to estimate parameters, but also to figure the likely size of the chance errors in the estimates.

5. The standard errors for *cluster samples* can be obtained by the *half-sample method*, splitting the sample into two halves and seeing how well they agree.

6. The formulas for the standard error have to take into account the details of the probability method used to draw the sample. The formulas which apply to simple random samples will usually underestimate the standard errors in cluster samples.

7. When bias operates more or less evenly across the sample, it cannot be detected just by looking at the sample data. Standard errors ignore that kind of bias.

8. The Current Population Survey, like all surveys, is subject to a number of small biases. The bias in the estimate of the unemployment rate is thought to be larger than the standard error.